

CSPG4 RNA across sarcoma types with an expanded comparison of extraskeletal myxoid chondrosarcoma

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Running title. Comparative CSPG4 RNA in EMC

Abstract

Background: Extraskeletal myxoid chondrosarcoma (EMC) is a rare soft-tissue sarcoma. Chondroitin sulfate proteoglycan 4 (CSPG4) is a cell-surface protein whose expression has been studied across sarcoma types. This study examined where EMC lies within a broader tissue RNA landscape and whether its differences from named sarcomas recur across datasets.

Methods: An exploratory analysis of a fixed 11-gene panel compared nine EMC specimens with 393 other specimens from 28 source-defined malignant tumor categories in one RNA-sequencing cohort. All 644 eligible specimens, including other tumor categories, remained in the descriptive atlas. Analysis used an exact rank-sum permutation test with Bonferroni correction for 11 genes, evaluated the array comparisons, and characterized additional public collections separately by source and measurement scale.

Results: Median CSPG4 was 54.18 TPM in EMC and 11.87 TPM in the defined malignant reference. The probability of higher EMC expression, allowing half credit for ties, was 0.846; exact two-sided $P=0.000130$ and adjusted $P=0.00143$. CSPG4 was higher than low-grade fibromyxoid sarcoma in both RNA sequencing and arrays, with comparison scores of 0.966 and 1.000. Gastrointestinal stromal tumors and dermatofibrosarcoma protuberans had slightly higher medians than EMC in the same RNA-sequencing source. External collections expanded histologic context but supplied no additional explicitly labeled EMC comparison.

Conclusions: These data place EMC toward the higher end of CSPG4 RNA expression within a defined sarcoma reference, with substantial overlap and comparator dependence. They support a broader comparative RNA account, without establishing diagnostic specificity, malignant-cell protein expression or therapeutic suitability.

Keywords. extraskeletal myxoid chondrosarcoma; CSPG4; sarcoma; RNA sequencing; public data; comparative expression

Introduction

Extraskelatal myxoid chondrosarcoma (EMC) is a rare soft-tissue sarcoma characteristically associated with NR4A3 rearrangements.[1] Public expression studies contain few EMC specimens, making it difficult to place candidate antigens in the context of other sarcomas.

The initial analysis compared RNA expression of 11 genes encoding candidate therapeutic antigens in EMC and selected sarcoma comparators. CSPG4 alone met the fixed prioritization rule across myxoid liposarcoma, low-grade fibromyxoid sarcoma and synovial sarcoma. Its higher expression relative to low-grade fibromyxoid sarcoma also recurred in a separate array collection. These findings motivated the broader question addressed here: where does EMC lie within the range of CSPG4 expression across sarcoma types?

CSPG4 encodes a cell-surface proteoglycan, also known as NG2, with an established role in experimental cancer biology. In melanoma cells, CSPG4 enhances integrin-dependent cell spreading and activation of focal adhesion kinase (FAK) and ERK1/2 signaling.[2] Boudin and colleagues described heterogeneous CSPG4 expression and associations with prognosis and immune features across soft-tissue sarcomas.[3] This provides a biological context for examining the small EMC groups available in public RNA-sequencing and array collections.[4,5]

The present study compares EMC with a broader sarcoma reference, examines contrasts shared across assays and describes additional public collections separately by source. Together, these analyses show which expression differences recur and which depend on the chosen comparator.

Methods

Study design and reference groups

CSPG4 was evaluated within a fixed panel of 11 genes: CD276, SSTR2, PRAME, FAP, CD248, CSPG4, MSLN, L1CAM, GPC3, ALPP and CDH17. The broader reference and uncertainty analyses were specified after the initial comparator-specific results were known. Supplementary Information describes the analysis chronology, specimen exclusions, probe mappings and comparator-specific decision rule.

The Hofvander collection provides gene-level transcripts per million (TPM), original diagnoses and specimen metadata.[4] Eligibility followed the original diagnoses and excluded specimens with a nonblank recurrence or metastasis flag. Of 13 source EMC specimens, three previously reported cases and one local recurrence were excluded, leaving nine. Source Table S1 marked all nine as fusion-positive. Filtered fusion records specify seven NR4A3 fusions, one RT-PCR-verified

FUS::NR4A2 fusion and no crosswalked fusion for the remaining specimen (Table S6). Membership follows the source EMC diagnosis.

The retained collection contains 644 specimens across 60 original diagnosis labels. The primary reference comprises 393 non-EMC specimens from 28 categories classified as Malignant by the source, excluding melanoma and unclassified spindle-cell tumors. A broader reference adds 96 specimens from nine Intermediate categories, including gastrointestinal stromal tumor (GIST) and dermatofibrosarcoma protuberans (DFSP), for 489 references across 37 labels. All eligible specimens remain in the descriptive atlas.

GSE24369 provides six EMC biopsies, 17 low-grade fibromyxoid sarcomas (LGFMS), six myxofibrosarcomas (MFS), six desmoids, five solitary fibrous tumors (SFT) and two pooled muscle RNAs.[5] Array analysis retained the released GPL6244 robust multi-array average log2 signals and CSPG4 transcript cluster 7990545. Lesion stage was not established. Comparisons were calculated within each dataset, keeping TPM and array signals separate.

Additional public collections

The source search, completed on 19 September, examined public sarcoma compendia, their original accessions and identifiable EMC-containing studies. Records required a usable CSPG4 measurement and traceable diagnosis. Each collection retained its source-specific measurement scale: Treehouse Tumor Compendium 25.01 log2(TPM+1), Boudin et al. published normalized log2 values, GSE213065 Salmon TPM, GSE234092 featureCounts-derived counts per million (CPM), and OpenPedCan v15 gene TPM.[3,6,7,8,9] The supplement lists unresolved or inaccessible EMC sources.

Selection used diagnosis and identity metadata rather than CSPG4 values. Treehouse profiles were grouped by donor identifier or conservative specimen prefix; Boudin records were joined to original GEO metadata. TCGA records already represented in Treehouse were omitted from Boudin summaries. GSE213065 contributed one earliest-timepoint T1a profile per explicit patient ID; GSE234092 retained its 69 repository records as a separately defined collection. OpenPedCan used independent-specimen lists, excluded known Treehouse donor aliases and kept one provenance-uncertain parental tumor outside clinical summaries. Analyses retained library-method and original-source strata. Counts describe profiles within resources, not a globally unique patient cohort; detailed selection and overlap ledgers are supplied.

Statistical analysis

Expression summaries included medians, quartiles and ranges. The comparison score $A = P(\text{EMC} > \text{reference}) + 0.5P(\text{tie})$ averages all specimen pairs; $A = 0.5$ denotes neutral ordering. The primary estimate weights reference specimens equally. Sensitivity analyses averaged the category-specific scores to give each reference category equal weight and examined deletion of individual EMC specimens.

The primary test used the exact rank-sum distribution for every allocation of nine EMC labels among 402 observations, preserving ties. Its two-sided tail includes sums at least as far from the null expectation as the observed sum. This tests an exchangeable-label, common-distribution null for the observational groups. Bonferroni correction covers the retained 11 genes. Approximate 95% percentile bootstrap intervals for both references used 50,000 independent resamples of specimens within each group at its original size, seed 20260921, with linear-interpolation quantiles.[10] No new hypothesis test was added for the broader reference.

Comparator-specific and sequencing-year-conditioned analyses were evaluated as sensitivities. External summaries and pairwise A values were calculated within designated source strata. Cross-source descriptive ranges included only within-source contrasts with at least five profiles per diagnosis in each contributing stratum. External measurements were not pooled, and no external P values were calculated. Code, source digests, selection ledgers and complete results accompany the data archive.

Results

EMC within the broader reference

Median CSPG4 was 54.18 TPM in the nine EMC specimens, with an interquartile range of 41.41–63.98 and a range of 10.14–207.08 TPM. The primary reference median was 11.87 TPM. The specimen-weighted A was 0.846 (approximate 95% bootstrap interval 0.709–0.958), with exact two-sided $P = 0.000130435$ and Bonferroni-11 $P = 0.00143479$. Equal weighting of reference categories gave $A = 0.866$. Deleting any one EMC specimen left A between 0.828 and 0.894.

EMC had the highest observed median among the 29 labels in the primary comparison, although distributions overlapped and several groups were small (Figure 1). GIST ($n = 14$) and DFSP ($n = 10$), both source-classified Intermediate, had slightly higher medians of 59.24 and 59.10 TPM. Adding all eligible Intermediate categories gave a reference median of 13.20 TPM and $A = 0.824$ (approximate 95% interval 0.678–0.943); equal-category A was 0.848. Several benign categories also had high values (Figure S1). Table S3 supplies every eligible diagnosis label.

MSLN also had higher ordering against the primary reference ($A=0.811$; adjusted $P=0.00406$), while several panel genes had lower ordering. Table S2 reports all 11 genes. These pooled comparisons answer a different question from the original comparator-specific rule, which CSPG4 alone satisfied.

Direct comparisons across assays

CSPG4 was higher in EMC than LGFMS in both the six-EMC versus 17-LGFMS array comparison ($A=1.000$) and the nine-EMC versus 13-LGFMS RNA-sequencing comparison ($A=0.966$; Figure 2). Deleting any one EMC or LGFMS array biopsy retained $A=1.000$. Array scores against MFS, SFT and desmoid were also 1.000; corresponding RNA-sequencing scores were 0.854, 0.879 and 1.000. DFSP supplied a contrary RNA-sequencing comparison ($A=0.467$).

Patterns in additional collections

The external characterization retained 1,334 Treehouse profiles, 986 Boudin records, 54 GSE213065 profiles, 69 GSE234092 records and 28 OpenPedCan clinical-context profiles. These supplied 304 source-by-diagnosis summaries and 1,480 within-source comparisons, but no additional explicitly labeled EMC group.

Some relative patterns recurred across sources (Figure S2). MFS was lower than leiomyosarcoma (LMS) in all six strata with at least five profiles per group, with $A=P(\text{MFS}>\text{LMS})+0.5P(\text{tie})$ ranging from 0.182 to 0.246. Undifferentiated pleomorphic sarcoma was lower than LMS in all five supported strata ($A=0.077\text{--}0.414$). Other comparisons varied: dedifferentiated liposarcoma had $A=0.173\text{--}0.343$ against LMS in four strata but 0.517 in the Nakayama cohort. GIST had $A=0.836$ against LMS in GSE234092 and 0.502 in Treehouse-processed SRP057793. The complete tables retain each comparison's source, scale and specimen counts.

Less favorable conditional sensitivities

The original three-histology CSPG4 composite was 0.89454 across all years and 0.81111 when conditioned on sequencing year. Only four distinct EMC patients supported the latter. Removing the 2019 sample group left two matchable EMC patients and a conditional score of 0.43333, while the all-year score remained 0.83333. The expanded-reference year-conditioned sensitivity gave $P=0.02064$ and adjusted $P=0.22704$, using a different statistic and 319 supported-year references. These sensitivities change the supporting patients and comparison sets; they do not establish a sequencing-year effect. Full conditional results and the separate CHRNA6, normal-tissue and PRAME evidence are retained in the supplement.

Discussion

EMC lay toward the higher end of CSPG4 RNA expression in both the primary malignant reference and the broader reference including intermediate tumors. Four named contrasts recurred across RNA sequencing and arrays. The slightly higher medians in GIST and DFSP show that elevated CSPG4 RNA extends across distinct mesenchymal tumor types. These groups provide useful comparators for subsequent EMC tissue studies, while the high values in some benign categories make cellular localization particularly important. Similar bulk expression does not establish a shared biological driver.

CSPG4 is already being investigated as a therapeutic target. A CSPG4-directed antibody inhibited tumor growth and metastasis in mouse models of triple-negative breast cancer.[11] In sarcoma models, cytokine-induced killer cells carrying a CSPG4-directed chimeric antigen receptor (CAR) showed activity in culture and in xenografts of leiomyosarcoma, undifferentiated pleomorphic sarcoma and fibrosarcoma.[12] A CSPG4-directed CAR-T approach has also entered phase I testing in recurrent or refractory head-and-neck squamous-cell carcinoma (NCT06096038), with safety and tolerability as the primary objective.[13] These studies place the EMC expression finding within an existing experimental targeting field, without establishing a treatment for EMC.

The contribution here is the explicit placement of EMC within the wider CSPG4 RNA distribution. The additional collections show both recurring and divergent relationships among other sarcomas, consistent with the heterogeneity reported by Boudin et al.[3] They supplied no further explicitly labeled EMC group, leaving independent replication of the EMC comparison unresolved.

The nine EMC specimens remain the main limit on precision, reflected in the wide bootstrap intervals. Small comparison groups also make median rankings uncertain. The references reflect source classifications and sampling rather than population incidence or a comprehensive clinical taxonomy. Equal-category estimates change the weights but retain this dependence. The broader exploratory comparison followed the identification of CSPG4 as a gene of interest. Correction for 11 genes addresses that family of tests; retrospective selection and unmeasured confounding remain outside both that correction and the approximate bootstrap intervals.

Incomplete specimen identities, clinical histories and historical overlap further limit interpretation. Assays and collections differ in quantification, library construction, disease stage and tissue composition, so cross-source measurements were kept separate. Uniform computational processing alone does not make them directly comparable. Array lesion stage and independence from historical collections remain incompletely established. The bounded source search and unresolved collections are documented in the supplement.

Finally, bulk RNA measures contributions from malignant and nonmalignant cells. Purity estimates were unavailable for all nine EMC specimens, and normal tissues were not matched to tumors. These data leave malignant-cell localization, accessible CSPG4 protein and normal-tissue sparing unresolved. Expression ordering alone cannot establish a diagnostic threshold, therapeutic window or treatment benefit. The next step is to establish CSPG4 protein abundance and cellular localization in EMC, including its accessibility on malignant cells. That would connect the RNA finding to the biological and targeting questions already being studied in other cancers.

Data code and declarations

Data availability. The original Hofvander data are available through <https://doi.org/10.5281/zenodo.17866629>. Original array data and annotations are available as GEO GSE24369 and GPL6244. Additional inputs are Treehouse Tumor Compendium 25.01, Boudin et al. Supplementary Table S8, GEO GSE213065 and GSE234092, and OpenPedCan v15. The data and code package, including exact source URLs, selection ledgers, digests and specimen values, is available at <https://github.com/trimcrae/Rare-cancers/blob/14fea9109d7f329fe9319f1af58560707c938027/research/release-candidates/cspg4-concise-20260922/journal-package/CSPG4-data-code.zip>. Source-specific attribution and reuse terms continue to apply.

Research scope. This study reanalyzes public data and includes no newly collected participant specimens or intervention.

CRedit authorship contribution statement. Tristan D. McRae: Project administration, Supervision. The author is responsible for the final interpretation and manuscript.

Funding. This research received no funding.

Competing interests. No competing financial interests exist. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. OpenAI Codex assisted with source organization, code implementation, computation, verification and drafting. The author takes responsibility for the article. Data visualizations were generated programmatically from the supplied measurements and estimates with AI-assisted code; no primary experimental image was generated or altered.

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CSPG4 RNA across the primary and broader reference categories

Burgundy: EMC Blue: primary Malignant reference Ochre: added Intermediate reference

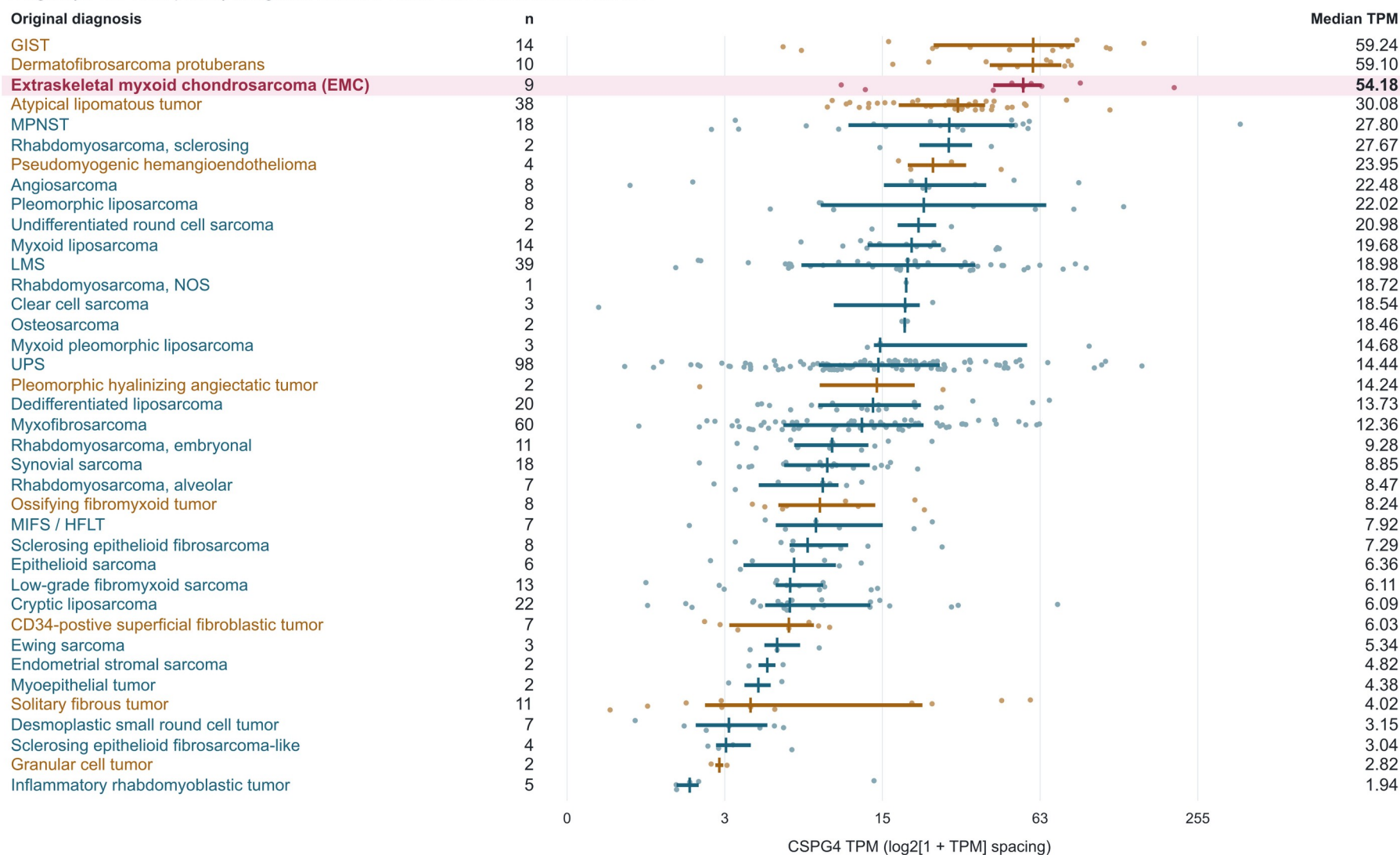


Figure 1. CSPG4 RNA in EMC and the primary and broader references. The 498 specimens span 38 source diagnosis labels: nine EMC (burgundy), 393 Malignant-category references (blue) and 96 additional Intermediate-category references (ochre). Dots show specimens, segments interquartile ranges and vertical marks medians. The axis uses $\log_2(1 + \text{TPM})$ spacing with TPM labels; categories are ordered by observed median. Other eligible contexts appear in Figure S1. MIFS/HFLT, myxoinflammatory fibroblastic sarcoma/hemosiderotic fibrolipomatous tumor; NOS, not otherwise specified.

Direct EMC contrasts shared across RNA sequencing and arrays

A = probability of higher EMC expression, with half-weight for ties

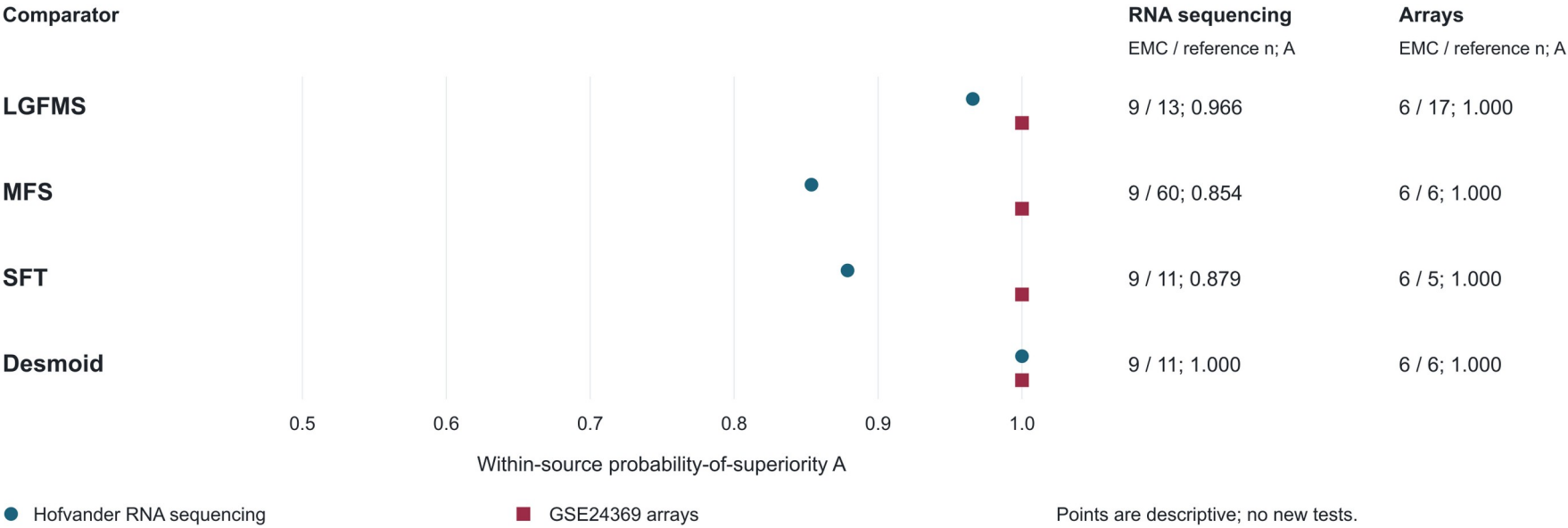


Figure 2. Direct EMC comparisons shared across two assays. Points show $A = P(\text{EMC} > \text{comparator}) + 0.5P(\text{tie})$ for the four diagnosis groups measured in both Hofvander RNA sequencing and GSE24369 arrays. The table supplies specimen counts and scores. Each score is calculated within its source. LGFMS, low-grade fibromyxoid sarcoma; MFS, myxofibrosarcoma; SFT, solitary fibrous tumor.